

An optical atomic clock based on a highly charged ion

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Steven A. King^{1,5,8}, Lukas J. Spieß^{1,8}✉, Peter Micke^{1,2,6}, Alexander Wilzewski¹, Tobias Leopold^{1,7}, Erik Benkler¹, Richard Lange¹, Nils Huntemann¹, Andrey Surzhykov^{1,3}, Vladimir A. Yerokhin¹, José R. Crespo López-Urrutia² & Piet O. Schmidt^{1,4}✉

Optical atomic clocks are the most accurate measurement devices ever constructed and have found many applications in fundamental science and technology^{1–3}. The use of highly charged ions (HCI) as a new class of references for highest-accuracy clocks and precision tests of fundamental physics^{4–11} has long been motivated by their extreme atomic properties and reduced sensitivity to perturbations from external electric and magnetic fields compared with singly charged ions or neutral atoms. Here we present the realization of this new class of clocks, based on an optical magnetic-dipole transition in Ar¹³⁺. Its comprehensively evaluated systematic frequency uncertainty of 2.2×10^{-17} is comparable with that of many optical clocks in operation. From clock comparisons, we improve by eight and nine orders of magnitude on the uncertainties for the absolute transition frequency¹² and isotope shift (⁴⁰Ar versus ³⁶Ar) (ref. ¹³), respectively. These measurements allow us to investigate the largely unexplored quantum electrodynamic (QED) nuclear recoil, presented as part of improved calculations of the isotope shift, which reduce the uncertainty of previous theory¹⁴ by a factor of three. This work establishes forbidden optical transitions in HCI as references for cutting-edge optical clocks and future high-sensitivity searches for physics beyond the standard model.

In HCI, the outer electron is strongly bound and thus experiences extremely magnified QED and nuclear size effects¹⁵. Binding energy, fine-structure and hyperfine splitting steeply scale up with the total charge of the ion. All this makes HCI highly sensitive devices for testing fundamental physical theories^{2,4}. Many of the atomic properties of HCI offer advantages for clock applications⁴. High binding energies shift electric-dipole-allowed (E1) transitions out of the optical range. This suppresses the sensitivity of the outer electrons to external electromagnetic perturbations^{5,8,10} by orders of magnitude in comparison with even the most insensitive species used in optical atomic clocks at present^{16,17}.

For the proof-of-principle demonstration of an HCI optical clock, we chose boron-like (B-like) Ar¹³⁺. With five bound electrons, its ground state ($1s^2 2s 2p_{1/2}$) is linked to the lowest excited state by a $^2P_{1/2} \rightarrow ^2P_{3/2}$ magnetic-dipole (M1) fine-structure transition with a wavelength of 441 nm. This transition is well suited for an initial demonstration, as it is the most precisely measured transition in any HCI. For much of the past two decades, the most precise measurements of optical transitions in HCI relied on the electron beam ion trap (EBIT) for HCI production and trapping. The use of grating spectrometers^{13,18,19} and laser spectroscopy²⁰ both afforded a Doppler-limited fractional uncertainty of 3×10^{-7} for Ar¹³⁺, limited by the high temperature ($T > 10^5$ K) of HCI trapped in EBITs. A recent experiment improved this by applying negative electronic feedback in a Penning trap for cooling Ar¹³⁺ down to the kelvin

level. Use of the continuous Stern–Gerlach effect for detection of the HCI's internal state after laser excitation yielded a fractional uncertainty of 9×10^{-9} (ref. ¹²). Still, an uncertainty gap of ten orders of magnitude persisted in comparison with state-of-the-art optical clocks²¹.

Recently, new techniques have overcome several key hindrances for HCI-based clocks: sympathetic cooling of hot HCI from megakelvin down to millikelvin temperatures using Be⁺ ions in a Paul trap²²; read-out of the HCI internal state by quantum logic spectroscopy (QLS)²³; and full quantum control over the motional state of a trapped HCI²⁴.

These breakthroughs, combined with the techniques and characterization presented here, have now enabled us to realize an HCI-based clock for the first time. Comprehensive measurements and analysis of all the systematic shifts of the clock transition result in a total fractional systematic uncertainty of 2.2×10^{-17} , with most shifts being below the 10^{-18} level. We determined the absolute frequency of the transition with a total fractional uncertainty of 1.5×10^{-16} by comparing it against a local optical clock using a single ¹⁷¹Yb⁺ ion and by using a previous measurement of its absolute frequency²⁵. Measurements performed using ⁴⁰Ar¹³⁺ and ³⁶Ar¹³⁺ yielded the isotope shift, with an uncertainty that shows QED nuclear recoil effects for the first time in a many-electron system. We therefore present improved calculations of the shift that reduce the uncertainty by a factor of three over previous work¹⁴, showing excellent agreement with the experimental value. Moreover, we extract the Landé *g*-factor of the $^2P_{3/2}$ state with higher accuracy than in earlier

¹Physikalisch-Technische Bundesanstalt, Braunschweig, Germany. ²Max-Planck-Institut für Kernphysik, Heidelberg, Germany. ³Technische Universität Braunschweig, Braunschweig, Germany.

⁴Institut für Quantenoptik, Hannover, Germany. ⁵Present address: Oxford Ionics, Begbroke, UK. ⁶Present address: CERN, Geneva, Switzerland. ⁷Present address: LPKF Laser & Electronics AG, Garbsen, Germany. ⁸These authors contributed equally: Steven A. King and Lukas J. Spieß. ✉e-mail: lukas.spiess@quantummetrology.de; piet.schmidt@quantummetrology.de

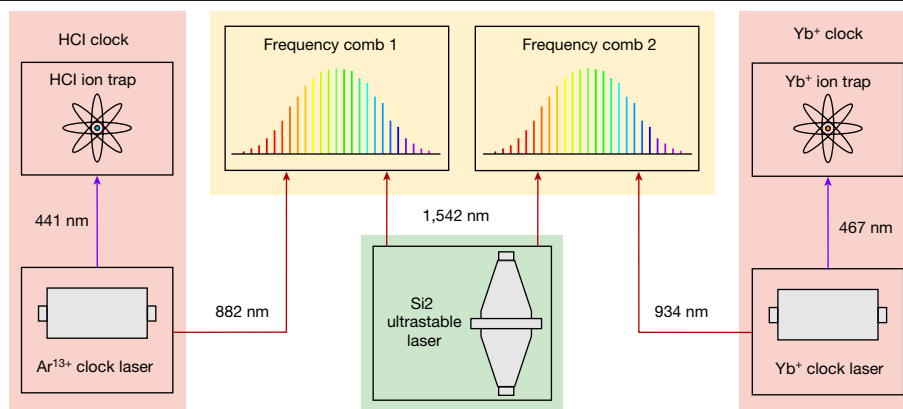


Fig. 1 | Scheme of the method for optical frequency comparison. Each of the two clock lasers (Ar^{13+} and $^{171}\text{Yb}^{+}$) is locked for pre-stabilization to its own local cavity and frequency comb, and ultimately steered to the corresponding optical transition by a digital control loop. The two frequency combs are

locked to the exceptionally stable cryogenic silicon cavity Si2 (ref. ³⁴). This method yields for each comb the frequency ratio between its clock laser and the Si2-stabilized laser. The dedicated laboratories are linked through phase-stabilized optical fibres⁵².

work²³ and establish an upper bound for the quadrupole moment of Ar^{13+} that confirms reported calculations^{26–29}. We also identify in this proof-of-principle experiment that an improved ion trap will bring the total systematic uncertainty below the 10^{-18} level.

Experimental setup

HCl are produced in a miniature EBIT³⁰ and transferred through an electrodynamic beamline to a linear Paul trap held at a temperature near 4 K (refs. ^{31,32}). There, a two-ion crystal composed of a single Ar^{13+} together with a single $^9\text{Be}^{+}$ ion is prepared. The latter is used for sympathetic cooling to below the millikelvin level and determination of the internal state of the HCl by QLS^{23,33}. In the trap, we reach a residual gas pressure below 10^{-12} Pa, suppressing charge-exchange collisions and enabling a mean HCl storage time of around 100 min.

The light for interrogation of the Ar^{13+} clock transition is based on an external-cavity diode laser with a wavelength of 882 nm. We pre-stabilize it with a local reference cavity and further by an optical frequency comb (see Fig. 1) by means of a frequency-steering acousto-optical modulator (AOM). The frequency comb is locked using a 1.5- μm reference laser to the cryogenic silicon cavity Si2 (ref. ³⁴), resulting in a fractional frequency stability of 4×10^{-17} at timescales of 1–100 s. The clock output is generated by adjusting the offset between the frequency comb and the pre-stabilized laser. Frequency doubling of the 882-nm light then produces the 441-nm light for driving the clock transition (for details, see Methods).

The clock cycle begins with a 38-ms-long sequence of laser pulses that performs the following steps. First, the ion crystal is cooled close to the motional ground state of the two axial normal modes using resolved-sideband cooling on the Be^{+} ion. The internal state of the HCl is then prepared using quantum-logic-assisted optical pumping²³, followed by algorithmic cooling of the two radial normal modes in which the HCl has a substantial amplitude of motion²⁴.

After this preparation sequence, the clock transition is interrogated using a 15-ms-long pulse from the 441-nm laser, which leads to an interaction-time-limited linewidth of approximately 50 Hz. This interrogation time is approximately 1.5 times the natural lifetime of the excited electronic state³⁵ and leads to the lowest achievable statistical uncertainty for our experimental parameters³⁶.

We use QLS to map the internal state information of the HCl after interrogation onto the logic ion Be^{+} , which is detected using the fast cycling transition³³. Using the result of repeated interrogations, the offset between the clock output and the line centre of any particular Zeeman component of the clock transition is steered with a dedicated

second-order integrating digital control loop (in the following, dubbed ‘servo’)³⁶.

Gradual charging of surfaces close to the trapped HCl by the ultra-violet laser beams used for the cooling and manipulation of the Be^{+} ion can induce slow position drifts of the ion, leading to a first-order Doppler shift³⁷. We suppress it by averaging the servo outputs for two counter-propagating clock beams for each Zeeman component. Consequently, eight independent servos run in parallel, two for each of the four Zeeman components shown in Fig. 2. As described in the Supplementary Material, we suppress further small systematic drifts by randomizing the order in which the interrogations for the eight servos are performed. The average of the eight servo outputs is free of the linear Zeeman shift, first-order Doppler shift and quadrupole shift³⁸ and steers the clock output. Its frequency is then directly compared with that of another clock by means of the optical frequency comb setup shown in Fig. 1, with only minor corrections necessary for those systematic shifts that are not averaged to zero.

As well as the dominant ^{40}Ar isotope, we are able to run the clock with the rare ^{36}Ar isotope. Details on its preparation are given in Methods.

Frequency measurement

Ar^{13+} features a comparatively broad natural linewidth of 17 Hz for the clock transition, leading to higher-frequency instability compared with other optical clocks. Nevertheless, it benefits from many of the

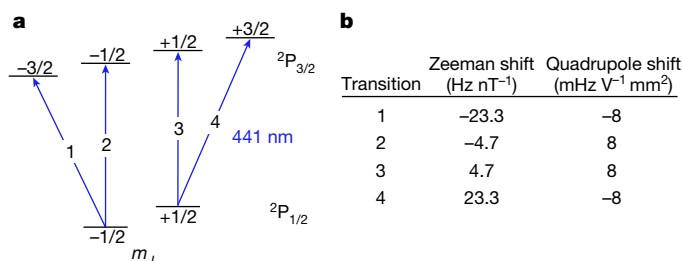


Fig. 2 | Zeeman components used in this work for the Ar^{13+} clock. **a**, Simplified term scheme of the transitions with magnetic quantum numbers (m_J) of their ground and excited states. **b**, Corresponding magnetic sensitivities (Zeeman shift) and maximum sensitivities to electric-field gradients (quadrupole shift)²⁸. During operation, a static magnetic field of 23 μT is applied for defining a quantization axis. The high charge state of the Ar^{13+} ion allows our trap to work with a gradient of approximately 2 V mm⁻², much lower than those needed in other linear ion traps used for optical clocks.

Table 1 | Systematic shifts ($\Delta\nu$) and corresponding 1- σ uncertainties (σ)

Shift	$\Delta\nu/\nu(10^{-17})$		$\sigma/\nu(10^{-17})$	
	$^{40}\text{Ar}^{13+}$	$^{36}\text{Ar}^{13+}$	$^{40}\text{Ar}^{13+}$	$^{36}\text{Ar}^{13+}$
Excess micromotion	-44.3	-55.0	2.2	2.1
Laser-induced a.c. Zeeman	0		0.2	
Secular motion	-0.1		<0.1	
Quadrupole shift	0		<0.1	
Quadratic Zeeman	<0.1		$\ll 0.1$	
Total Ar^{13+} clock	-44.4	-55.1	2.2	2.1
Statistics	0		10	13
Uncompensated paths	0		0.1	
$\text{Yb}^+ E3$ absolute frequency	0		13	
Gravitational redshift	16.4		0.6	
Total absolute frequency	-28	-39	15	17

Values are given relative to the transition frequency ν . When applicable, differing values for $^{40}\text{Ar}^{13+}$ and $^{36}\text{Ar}^{13+}$ are given. The different values for the excess micromotion shift are primarily caused by the difference in isotopic mass. The absolute frequencies are limited by the statistical uncertainties and the total uncertainty of the previous measurement of the absolute frequency of the $^{171}\text{Yb}^+$ clock transition, which contribute approximately equally. Further systematic shifts are at the 10^{-19} level or below (see Extended Data Table 2 and the Supplementary Material).

advantageous properties of HCl, such as a small differential polarizability and quadrupole moment. The dominant systematic shift in our system is caused by residual driven motion at the frequency of the ion trap's radially confining potential, referred to as excess micromotion, which leads to time dilation relative to the laboratory frame. In the present case, an unexpected component along the axial direction of the trap that cannot be compensated was probably caused by a misalignment of the trap electrodes³¹. Here it is by far the largest frequency shift and limits the relative uncertainty to 2.2×10^{-17} . However, this shift is not exacerbated by the large charge state of the HCl because it depends only on the secular frequency of the ion in the confining potential, which we chose to be similar to that used for singly charged ions. An improved ion trap will reduce this uncertainty to well below the 10^{-18} level³⁹. The next largest systematic uncertainty is at the low 10^{-18} level and arises from the a.c. Zeeman shift induced by the clock laser off-resonantly coupling to neighbouring Zeeman states. The uncertainties on all other investigated systematic shifts, including time-dilation shifts from residual thermal motion, shifts from the electric quadrupole moment of the excited $^2P_{3/2}$ clock state coupling to an electric-field gradient and higher-order magnetic field shifts, are even smaller under our operating conditions.

As reference for the frequency of the Ar^{13+} clock, an optical clock based on the $^2S_{1/2} \rightarrow ^2F_{7/2}$ electric octupole ($E3$) transition in $^{171}\text{Yb}^+$ is used. This system has a fractional systematic uncertainty of 3×10^{-18} and a fractional frequency instability of $1 \times 10^{-15} \tau^{-1/2}$, in which τ is the averaging time in seconds. Because the $^{171}\text{Yb}^+$ and Ar^{13+} ions are trapped at unequal heights, a correction for the difference in gravitational potential is applied.

The outputs of the $^{171}\text{Yb}^+$ and Ar^{13+} clocks were delivered to separate optical frequency combs through phase-stabilized optical fibre links, as shown in Fig. 1. Each comb allows a direct measurement of the local ratio between the frequencies of the clock and the ultrastable 1.5- μm laser³⁴. Combination of the synchronous measurements of the two ratios yielded the ratio between the two clock laser frequencies.

Table 1 summarizes the dominant systematic frequency shifts on the Ar^{13+} clock transition and their uncertainties, as well as those

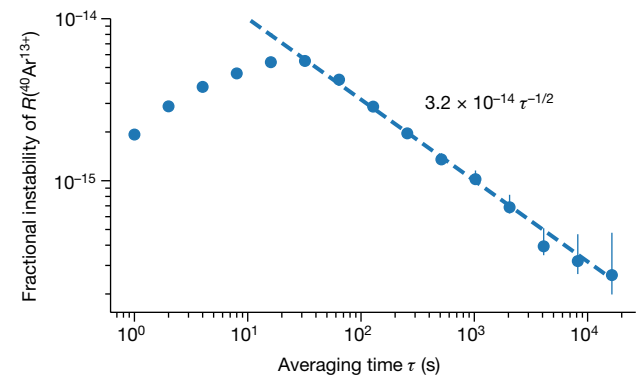


Fig. 3 | Instability of the ratio between the $^{40}\text{Ar}^{13+}$ and $^{171}\text{Yb}^+$ transition frequencies. Statistical uncertainty of the ratio $R(^{40}\text{Ar}^{13+}) = \nu(^{40}\text{Ar}^{13+})/\nu(^{171}\text{Yb}^+ E3)$ as a function of averaging time τ , as measured using the overlapping Allan deviation. The error bars correspond to the 68% confidence interval. The instability improves as $\tau^{-1/2}$ beyond the servo time constant of around 30 s, as characteristic for white frequency noise. The dashed line is a fit to the data for averaging times beyond 300 s.

arising from the measurement of the absolute transition frequency. An in-depth analysis of all of the systematic shifts pertinent to the presented measurements can be found in the Supplementary Material.

Results and analysis

The ratio between the Ar^{13+} and $^{171}\text{Yb}^+ E3$ transition frequencies were repeatedly measured over several days for each of the two Ar isotopes, with a total measurement time of approximately 100,000 s for $^{40}\text{Ar}^{13+}$ and 50,000 s for $^{36}\text{Ar}^{13+}$. The frequency instability of the complete dataset for $^{40}\text{Ar}^{13+}$ is shown in Fig. 3. At timescales beyond the servo time constant of 30 s, we infer a frequency instability of $3.2 \times 10^{-14} \tau^{-1/2}$. For $^{36}\text{Ar}^{13+}$, a lower instability of $2.6 \times 10^{-14} \tau^{-1/2}$ was obtained, resulting from more efficient state preparation of the ions, reducing the dead time of the experimental cycle. We do not observe a floor in the frequency instability out to averaging times of more than 10,000 s, as expected for white frequency noise. Extrapolating the observed instability to the total measurement time resulted in a fractional statistical uncertainty of 1×10^{-16} for each of the two frequency ratios.

The final results are summarized in Extended Data Table 1. The absolute values of the $^{40}\text{Ar}^{13+}$ and $^{36}\text{Ar}^{13+}$ clock-transition frequencies are derived from the absolute frequency of the $^{171}\text{Yb}^+ E3$ transition reported in ref. ²⁵. The overall fractional frequency uncertainties of 1.5 and 1.7×10^{-16} , respectively, place these results among the most accurately reported absolute transition frequencies of all time^{25,40–42}. For $^{40}\text{Ar}^{13+}$, the value agrees well with the most recently published measurement¹², surpassing its accuracy by almost eight orders of magnitude.

Combining the data for the two isotopes allowed us to determine the $^{40}\text{Ar}^{13+}$ – $^{36}\text{Ar}^{13+}$ isotope shift with a fractional uncertainty of 5.7×10^{-11} , in which some systematic uncertainties were common-mode rejected. Our result confirms the only previous measurement¹³ but has an uncertainty that is almost nine orders of magnitude lower. This newly achieved level of uncertainty allows benchmarking of predictions from atomic-structure calculations. In particular, we are now able to resolve contributions from the QED nuclear recoil.

Our relativistic configuration-interaction (CI) calculation is supplemented by the separate treatment of the QED effects to the mass shift and to the energy levels. The QED calculation of the mass shift is carried out within the rigorous approach that goes beyond the external-field approach of QED and takes into account all orders in the electron–nucleus Coulomb interaction⁴³. Details about the theoretical calculations can be found in Methods and the Supplementary Material.

Table 2 | Theoretical and experimental values for the isotope shift

	This work	CI-DFS ¹⁴	MCDF ⁵¹
Mass shift, relativistic	1,906.45(80)	1,920	1,915
Mass shift, QED	−13.0(36)	−17(11)	
Field shift, relativistic	−15.9(15)	−16(1)	−16
Field shift, QED	−0.08		
Total theory	1,877.5(40)	1,886(11)	1,899
Experiment, this work	1,878.11053251(11)		
Experiment ¹³	1,895(93)		

The various contributions to the isotope shift $\nu(^{40}\text{Ar}^{13+}) - \nu(^{36}\text{Ar}^{13+})$ are discussed in the main text, with further details in Methods and the Supplementary Material. The total calculated shift is in excellent agreement with the experimental value. All shifts are given in MHz. CI-DFS, configuration-interaction Dirac–Fock–Sturm; MCDF, multiconfiguration Dirac–Fock.

Theoretical results for the individual effects contributing to the isotope shift are presented in Table 2. It is seen that our calculations are in good agreement with but more accurate than the previous computations reported in the literature. The accuracy of the total theoretical value for the isotope shift was improved by a factor of three as compared with the previous best calculation¹⁴. We observe that the QED part of the mass shift is comparable with the field shift and thus cannot be neglected in the theoretical description of the isotope shift, as has been customary in the literature previously. The experimental result is in excellent agreement with the theoretical prediction. It should be mentioned that the calculations and experiments were carried out blind to one another, preventing any bias. The comparison of the experimental and theoretical results tests the QED mass shift on the level of 31%. To the best of our knowledge, this is the first experimental verification of the QED nuclear recoil effect in a many-electron system, as all previous tests of QED were performed within the static external-field regime.

Data taken during the clock comparisons allow the determination of atomic properties as described in Methods. We obtain a quadrupole moment of the $^2P_{3/2}$ state of $-0.02(9) ea_0^2$ for $^{40}\text{Ar}^{13+}$ and $-0.03(14) ea_0^2$ for $^{36}\text{Ar}^{13+}$, in which e is the elementary charge and a_0 is the Bohr radius. Both results confirm the predicted small value of $0.0235 ea_0^2$ (refs. ^{28,29}). The ratio between the g -factors of the excited and ground states $r = g_{3/2}/g_{1/2}$ was determined to be $r(^{40}\text{Ar}^{13+}) = 2.0075142(25)$ for $^{40}\text{Ar}^{13+}$. With the previously measured $g_{1/2}$ from ref. ⁴⁴, which has parts-per-billion uncertainty, we obtain $g_{3/2} = 1.3322837(17)$, being in good agreement with our previous measurement²³. The fourfold reduced uncertainty achieved here brings it to the level of a theoretical prediction of $g_{3/2}^{\text{th}} = 1.332\,282\,5(14)$ and confirms it²⁷.

For $^{36}\text{Ar}^{13+}$, we determine a g -factor ratio of $r(^{36}\text{Ar}^{13+}) = 2.0075154(32)$, which is not substantially different from the $r(^{40}\text{Ar}^{13+})$ result. No precise experimental value is available for $g_{1/2}$ for this isotope. The difference $r(^{36}\text{Ar}^{13+}) - r(^{40}\text{Ar}^{13+}) = 1.2(36) \times 10^{-6}$ is consistent with the theoretical prediction of 2.2×10^{-6} (refs. ^{26,27}).

Conclusions

The first operation and evaluation of an optical atomic clock based on a highly charged ion was presented. The achieved performance confirms the applicability of HCI in optical clocks. The frequency instability of the HCI clock can be greatly improved by choosing an HCI species with a longer excited-state lifetime, such as Ni^{12+} , Pd^{12+} or Pr^{9+} (refs. ^{45,46}). This will allow HCI-based clocks to compete with the best available optical atomic clocks.

The potential of HCI to contribute to fundamental physics was demonstrated by experimentally confirming QED nuclear recoil contributions to the isotope shift between $^{40}\text{Ar}^{13+}$ and $^{36}\text{Ar}^{13+}$. This was possible because the properties of few-electron HCI are well calculable using

ab initio atomic-structure methods. Following this work, isotope shift measurements of optical-clock transitions can now be extended to very different charge states of the same element and thus contribute to the search for hypothetical fifth forces^{4,47,48}. Frequency comparisons between HCI clocks and other optical clocks as demonstrated here can also be used to look for beyond-the-standard-model physics. In particular, HCI optical-clock transitions have the largest known sensitivity to both a change in the fine-structure constant and ultralight scalar dark matter of all known atomic systems^{2,5–7,49,50}.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-022-05245-4>.

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Loading of the different isotopes

Highly charged argon ions are produced in a miniature EBIT³⁰ by injecting gas (here ³⁶Ar or ⁴⁰Ar) through a differentially pumped injection system into the trap region, in which the atoms are ionized by the electron beam. ⁴⁰Ar has a natural abundance of 99.6% and hence no further purification stages are required within the EBIT and beamline when working with this isotope. The second most abundant isotope is ³⁶Ar, with a natural abundance of only 0.33%. We use an enriched sample of this isotope with close to 100% purity. To minimize the necessary quantity of the enriched gas, a dedicated injection system with small dead volume was built.

The EBIT operating parameters and breeding time are optimized for maximum yield of Ar¹³⁺. After a short breeding time of a few hundred milliseconds, the HCl are ejected from the EBIT and launched into an ion-optical beamline connecting it to the Paul trap^{22,23}. The velocities of the different charge states and hence their times of flight through the beamline depend on their charge-to-mass ratios. We filter out unwanted charge states in the beamline with an ion-optical element referred to as a ‘gate’ electrode normally held at a high potential to reject the HCl. It is pulsed to a lower value at the arrival time of the desired charge state, allowing it to pass. The lower mass of ³⁶Ar necessitates minor changes to the switching times of the beamline electrodes compared with ⁴⁰Ar¹³⁺.

Clock and logic lasers

A small fraction of the 441-nm light generated by frequency doubling of the 882-nm laser light is used to injection lock a Fabry–Pérot laser diode with a free-running wavelength of 440 nm at 30 °C, which provides the higher-power beams required for fast quantum-logic and algorithmic-cooling²⁴ operations. The entire clock and logic laser setup is enclosed in a housing to avoid air-flow and temperature variations, allowing operation over several days without adjustments.

The clock laser beams are delivered in the radial direction of the trap, perpendicular to the symmetry axis. This has two advantages over our original configuration²³, in which the beam entered along the symmetry axis: first, the motional modes that the laser can be coupled to have a higher frequency and thus the motional sidebands are better separated from the carrier transition, and yield a smaller Lamb–Dicke parameter. This suppresses line-pulling effects on the carrier from the sideband transitions. Second, it is desirable to alternate the direction of interrogation between two counter-propagating directions to eliminate any potential first-order Doppler shift. This is more difficult along the trap axis owing to the presence of the 1.2-m-long ion-optical beamline. The QLS beam still enters along the trap axis to ensure strong coupling to the axial sidebands used in this method.

The average of the frequencies of the eight servos steer the clock output by adjusting the offset of the clock laser to the optical frequency comb mode. This decouples the clock laser from the ultrastable 1.5-μm laser on timescales beyond the servo time constant and ensures that the laser frequency measured using the comb is a real-time reflection of the transition frequency with the linear Zeeman, linear Doppler and quadrupole shifts removed. Only a constant frequency offset remains, which depends on the drive frequencies of the various AOMs. This procedure makes it unnecessary to carry out synchronized measurements of several radiofrequency signals, such as AOM drive frequencies, in the various laboratories.

Quadrupole moment of the ²P_{3/2} state

States with $J \geq 1$, such as the excited ²P_{3/2} state in Ar¹³⁺, possess electric quadrupole moments Θ . Electric-field gradients experienced by the HCl will therefore lead to quadrupole shifts of its sub-levels that depend on $|m_J|^2$ (ref. ⁵³). For a two-ion crystal aligned along the symmetry axis of a linear Paul trap, the d.c. potential U used for axial confinement has a gradient at the position of the HCl. For our trap geometry, the axial potential is described by:

$$U = \frac{1}{2} \frac{dE_z}{dz} (x^2 - z^2) \quad (1)$$

in which dE_z/dz is the electric-field gradient along the axial symmetry axis z of the trap. The x axis is defined along the two radiofrequency blade electrodes³¹. The lack of y dependence is specific to the geometry of our trap and is confirmed by simulations and experiments. The gradient dE_z/dz can be determined by measuring the single-ion or two-ion motional frequencies in the z direction. A second contribution arising from the charge of the Be⁺ ion⁵⁴ is smaller by a factor of seven compared with that of the trapping field, owing to its large separation of 21 μm from the Ar¹³⁺ ion.

This electric field gradient of the trap shifts the Zeeman sub-levels of the excited state according to the relation

$$\Delta v_{qs} = \frac{\Theta}{4h} \frac{dE_z}{dz} \frac{J(J+1) - 3m_J^2}{J(2J-1)} (3 \cos^2 \beta - 1 - \epsilon \sin^2 \beta \cos(2\alpha)), \quad (2)$$

in which $\beta = 30(5)^\circ$ is the fixed angle between the electric-field gradient and the quantization axis in our setup, $\alpha = 45(5)^\circ$ is the angle between the x axis and the plane defined by the z axis and our quantization axis, and $\epsilon = 1$ accounts for the breaking of cylindrical symmetry. The shift is averaged in real time by choosing to operate with servos that sample all of the Zeeman substates of the excited state. Nevertheless, its value can be determined by comparing the mean frequency of pairs of Zeeman components that share the same value for $|m_J|^2$.

g -factor of the ²P_{3/2} state

The ratio r between the Landé g -factors of the excited $g_{3/2}$ and the $g_{1/2}$ ground states can be determined from the ratio ρ of the splitting between the outer and inner Zeeman component pairs:

$$r = \frac{g_{3/2}}{g_{1/2}} = \frac{1 - \rho}{3 - \rho}, \quad (3)$$

with $\rho = (v_4 - v_1)/(v_3 - v_2)$, in which the labelling of the four transitions follows that in Fig. 2. We calculate r from servo data on a cycle-by-cycle basis to avoid degradation of the measurement owing to drifts in the magnetic field during the measurement period.

The measured ratio r is also affected by the oscillating magnetic field from the radiofrequency drive of the trap, which couples the Zeeman substates of each manifold^{55,56}. Previously, its mean-square amplitude along the quantization axis of our trap has been bounded to be smaller than $3 \mu T^2$ at drive amplitudes twice as high as in this work³¹. For a conservative estimate of this contribution, we use this value and assume on the basis of the geometry of our trap that its transverse components have the same magnitude. This leads to a fractional shift bounded to $|\Delta r/r| < 7 \times 10^{-7}$, with this value taken fully as a further uncertainty.

Data processing

Spurious data were identified and removed in various ways. First, cycle slips on the tracking oscillators used by the Ar¹³⁺ experiment were checked by having two oscillators with asymmetrically detuned free-running frequencies and with slightly different loop gains tracking each beat signal. Both are counted and typically agree at the mHz level for sampling times of 1 s used for all frequency-counting processes. To detect the cycle slips, we apply a threshold value of 0.1-Hz difference between the two counters, as a cycle slip of 2π during the sampling time of 1 s would result in a 1-Hz jump. For the other beats involved in the frequency comparison, only a single counter channel was used. In these cases, deviations were found by applying a threshold of 0.1 Hz from the nominal, fixed beat frequency, which is between 5 and 100 times larger than the typical maximum fluctuations of the various beat frequencies. Second, any sudden glitches in the Ar¹³⁺ clock laser

frequency caused by a momentary loss of the phase lock to the laser stabilized to the Si2 cavity³⁴ were identified.

In all cases described above, the faulty data were removed, along with 30 s of data afterwards. Furthermore, 5 s of data was removed before the faulty period to allow for any offset in the synchronization between the frequency counters in the different laboratories.

Further data were excluded when one or more of the eight clock servos temporarily lost lock, for example after an unusually rapid change in the local magnetic field. Data in the vicinity of unusual changes in the magnetic field were removed, even if the servos remained in lock throughout.

To exclude possible errors in the data analysis, at least two independent blind evaluations of each dataset were carried out. In all cases, they agreed well within the measurement uncertainties.

Theory of the isotope shift

The isotope-dependent part of an atomic transition energy can be represented as a sum of the so-called mass-shift (E_{ms}) and field-shift (E_{fs}) terms,

$$E_{iso} = E_{ms} + E_{fs} = \frac{m_e}{M} K + \left(\frac{R}{\lambda_C} \right)^2 F, \quad (4)$$

in which m_e is the electron mass, M is the nuclear mass, $R = \langle r^2 \rangle^{1/2}$ is the root-mean-square radius of the nuclear charge distribution and $\lambda_C = \lambda_C / (2\pi)$ is the reduced Compton wavelength. K and F are the mass-shift and the field-shift constants, respectively; they depend on the electronic structure of the atom but not on the nuclear properties of the isotope.

The relativistic mass-shift constant is induced⁴³ by the expectation value of the relativistic recoil operator, $K = \langle \bar{H}_{rec} \rangle$, with

$$\bar{H}_{rec} = \frac{1}{2} \sum_{ik} \left[\mathbf{p}_i \cdot \mathbf{p}_k - \frac{Z\alpha}{r_i} \left(\boldsymbol{\alpha}_i + \frac{(\boldsymbol{\alpha}_i \cdot \mathbf{r}_i) \mathbf{r}_i}{r_i^2} \right) \cdot \mathbf{p}_k \right]. \quad (5)$$

Here the indices i and k numerate the electrons, $\mathbf{p}$ and $\mathbf{r}$ are the momentum and coordinate operators, respectively, and $\boldsymbol{\alpha}$ is the vector of Dirac matrices. The relativistic mass shift of the $^2P_{1/2} \rightarrow ^2P_{3/2}$ transition in B-like argon was calculated by the CI method, implemented in our previous works⁵⁷. In our calculation, we account for a strong mixing of the $1s^2 2s^2 2p$ states with the closely lying excited states by treating all possible $1s^2 2l' 2l''$ configurations as several reference states, in which l, l' and l'' denote the angular momenta of electron orbitals. The CI expansion of our large-scale computation included single, double, triple and quadruple excitations from the multiple reference states specified above.

To obtain accurate results for the mass-shift constant K , the relativistic CI treatment needs to be supplemented by a separate treatment of QED nuclear recoil effects. The underlying theory was developed in ref.⁴³. In this work, we calculate the QED recoil effect within the independent-electron approximation, for which the one-electron wave functions are obtained by solving the Dirac equation with a localized Dirac–Fock potential. Furthermore, we account for the correction arising from the QED shifts of energy levels. Because of a large mixing between closely lying states, the QED energy shifts alter the mixing coefficients and thus induce a notable contribution to the mass-shift constant. This correction was calculated by including the effective QED operator^{58,59} into the Hamiltonian matrix of the CI procedure.

$$F = \langle V_{fs} \rangle = \left\langle \sum_i \frac{dV_{nuc}(r_i)}{d(R/\lambda_C)^2} \right\rangle. \quad (6)$$

The relativistic value for the field-shift constant F is obtained as an expectation value of the derivative of the nuclear binding potential V_{nuc} over the square of the nuclear charge radius R^2 . We calculate the constant

F by averaging the operator V_{fs} with the CI wave functions obtained in the same way as for the mass shift. The field shift of the $^2P_{1/2} \rightarrow ^2P_{3/2}$ transition energy in Ar^{13+} is much smaller than the mass shift. The reason for this is that the overlap of the $2p$ one-electron orbitals with the nucleus is very small. Thus the dominant effect comes from mixing with the $1s$ core orbital, which is suppressed by the parameter $1/Z$. Because of its smallness, the field shift does not introduce any contribution to the uncertainty budget of the total isotope shift in the case under consideration.

The relativistic value of the field-shift constant needs to be supplemented by an estimate of the leading QED effects. We find that the leading contribution comes from the QED shifts of energy levels. This effect is enhanced owing to strong mixing of the reference states with the closely lying virtual excited states. We calculate the QED correction resulting from energy shifts by including the effective QED operator^{58,59} into the Hamiltonian matrix of the CI method and repeating our calculations with and without the QED addition. The second-largest QED effect is the pure radiative finite-nuclear-size correction. Following ref.¹⁴, we estimate it by rescaling the known hydrogenic result for the $1s$ state from ref.⁶⁰.

We summarize our calculations in Table 2. The nuclear parameters used in the calculation are: $M_{40}/m_e = 72,828.99631$, $M_{36}/m_e = 65,546.85208$, $R_{40} = 3.4274(26)$ fm and $R_{36} = 3.3905(23)$ fm. The charge radii are taken from ref.⁶¹. The quoted error of the theoretical prediction includes an uncertainty of 1.5 MHz owing to the experimental values of the nuclear radii. The nuclear masses are calculated by subtracting the mass of the electrons and binding energies from the atomic masses listed in ref.⁶². Further details about these calculations are given in the Supplementary Material.

Data availability

The traces of the frequency ratios are available at <https://doi.org/10.5281/zenodo.6901524>. Further datasets generated and analysed during this study are available from the corresponding author on request.

Code availability

All code that has been used to generate or analyse data during this study are available from the corresponding author on request.

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Article

Author contributions S.A.K., L.J.S., P.M., T.L., E.B., J.R.C.L.-U. and P.O.S. developed the experimental setup. S.A.K., L.J.S., P.M., A.W., R.L. and N.H. carried out the experiments. S.A.K., L.J.S., A.W. and E.B. analysed the data. J.R.C.L.-U. and P.O.S. conceived and supervised the study. A.S. and V.A.Y. performed the theoretical calculations. S.A.K., L.J.S., A.S. and P.O.S. wrote the initial manuscript, with contributions from P.M. and J.R.C.L.-U. All authors discussed the results and reviewed the manuscript.

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Additional information

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Correspondence and requests for materials should be addressed to Lukas J. Spieß or Piet O. Schmidt.

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Extended Data Table 1 | Measured frequency ratios and absolute frequencies

Measurement	Value	Relative uncertainty
$R(^{40}\text{Ar}^{13+})$	1.057 769 387 587 480 94(11)	1.0×10^{-16}
$\nu(^{40}\text{Ar}^{13+})$	679 216 462 397 957.43(11) Hz	1.5×10^{-16}
$R(^{36}\text{Ar}^{13+})$	1.057 766 462 735 187 48(13)	1.2×10^{-16}
$\nu(^{36}\text{Ar}^{13+})$	679 214 584 287 424.91(12) Hz	1.7×10^{-16}
$\nu(^{40}\text{Ar}^{13+}) - \nu(^{36}\text{Ar}^{13+})$	1 878 110 532.51(11) Hz	5.7×10^{-11}

Optical frequency ratios $R(^X\text{Ar}^{13+})=\nu(^X\text{Ar}^{13+})/\nu(^{171}\text{Yb}^+ E3)$, derived transition frequencies $\nu(^X\text{Ar}^{13+})$, resulting isotope shift $\nu(^{40}\text{Ar}^{13+})-\nu(^{36}\text{Ar}^{13+})$ and total relative uncertainties of each of the measurements are given.

Extended Data Table 2 | Investigated systematic shifts ($\Delta\nu$) and corresponding 1- σ uncertainties (σ) for the Ar^{13+} clock

Shift	$\Delta\nu/\nu$ (10^{-17})		σ/ν (10^{-17})	
	$^{40}\text{Ar}^{13+}$	$^{36}\text{Ar}^{13+}$	$^{40}\text{Ar}^{13+}$	$^{36}\text{Ar}^{13+}$
Excess micromotion	-44.3	-55.0	2.2	2.1
Laser-induced a.c. Zeeman	0		0.2	
Secular motion	-0.1		< 0.1	
Quadrupole shift	0		< 0.1	
First-order Doppler	< 0.1		< 0.1	
First-order Zeeman	0		< 0.1	
Quadratic Zeeman	< 0.1		$\ll$ 0.1	
Blackbody radiation	$\ll$ 0.1		$\ll$ 0.1	
Servo error	0		< 0.1	
AOM phase chirp	< 0.1		< 0.1	
Total	-44.4	-55.1	2.2	2.1

Values are given relative to the transition frequency ν . A detailed analysis is given in the Supplementary Material.